

Bhuvan Sarakam

Senior Data Engineer | AI/ML & Data Platform Engineering

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PROFESSIONAL SUMMARY

Senior Data Engineer with 10+ years of end-to-end experience across banking, insurance, credit risk, fraud, and asset management, spanning the full arc from hands-on AWS/Spark ETL engineering (Cognizant, 2016) through fraud and credit-risk modeling (American Express, Citi, TransUnion) to enterprise lakehouse and AI agent engineering at Janus Henderson today. Writes and tunes SQL, T-SQL, Snowflake SQL, and Spark SQL daily — stored procedures, views, MERGE-based upserts, and transaction-safe schema migrations — alongside Databricks/PySpark lakehouses, Snowflake warehouses, Azure Data Factory pipelines, and .NET services, with ownership of dimensional modelling, incremental load design, and serverless observability across Azure and AWS ecosystems. Builds production AI agent tooling (LangGraph, MCP, Claude/OpenAI/Gemini) that automates BAU triage, SQL-based data-integrity auditing, and pipeline monitoring.

TECHNICAL SKILLS

SQL & Database Engineering: SQL, T-SQL, Snowflake SQL, Spark SQL, Oracle SQL/PL-SQL — stored procedures, views, functions, triggers, joins, subqueries, CTEs, window functions, MERGE/upsert and SCD logic, query optimization and execution-plan tuning, transaction-safe/idempotent migrations, data-quality and reconciliation queries

Data Engineering: Databricks, Apache Spark (PySpark), Delta Lake, Azure Data Factory (ADF), ETL/ELT pipeline design, Kimball dimensional modelling, incremental loading, Airflow, Kafka, SSIS, dbt

Warehouse & Storage: Snowflake, SQL Server, Amazon Redshift, Cosmos DB, Amazon S3, Azure Blob, ADLS Gen2

Cloud: AWS (EC2, S3, RDS, Redshift, EMR, Step Functions, VPC, IAM, CloudWatch, CloudFormation), Azure (Functions, Synapse, Event Hubs, APIM, Bicep)

AI / GenAI: AI Agent Development, LLM Integration (Claude, OpenAI, Gemini), MCP (Model Context Protocol), LangChain, LangGraph, RAG Pipelines, MLflow

Backend & APIs: C#, .NET Core, ASP.NET Core, RESTful APIs, FastAPI, Azure API Management (APIM)

Languages: SQL, T-SQL, Python, PySpark, PowerShell, R, SAS

DevOps & BI: Azure DevOps YAML CI/CD, GitHub Actions, Git, Jira, Power BI, DAX, SSRS

BUSINESS IMPACT

- **Scale of delivery** — 398 authored commits across 14 DMDM delivery work streams and 3 production repositories, spanning a .NET application estate, a Databricks lakehouse and a Snowflake SQL warehouse over 4 release trains.
- **BAU and incident reduction** — investigated 9 distinct BAU and production-support problem themes and auto-triaged 50 production pipeline runs (10 failures, 40 SLA breaches, 25 classified SEV0) into 50 logged triage entries with owner, cause and first-line fix, cutting mean time to detect from hours to 5 minutes.
- **Data-integrity protection** — used SQL-based row/column diagnostics to identify 120 actionable split-HFD mappings and 184 missing downstream registration outputs affecting HEDS/JHIQ consumers, recovered 4 corrupted account cross-reference rows after a staging bulkload failure, and traced a concurrency defect where pipeline executions collided 13 seconds apart.
- **Career-long delivery track record** — 92% ML prediction accuracy vs. 78% baseline (Cognizant); 40% fraud-latency reduction at real-time scale (American Express); approval-rate lift from 51% to 90% via Decision Tree policy optimization (Citi); <10% quarterly error rate on CECL survival models (TransUnion); 35–40% ETL/data-latency reductions delivered at Zurich, Hubbell, and Janus Henderson.

PROFESSIONAL EXPERIENCE

Application Developer (Full Stack, Data, Python & AI/ML) | Janus Henderson Investors — Denver, CO **Sept 2024 – Present**

- Engineered a dimensional lakehouse on Databricks, as measured by 30 dimension, fact and bridge artifacts delivered across 158 commits and 77 PySpark notebooks, by authoring parameterised Apache Spark loaders and Spark SQL transformations for portfolio, product, share-class and product-group dimensions plus strategy and composite link facts.
- Optimized incremental load throughput by replacing full reloads with SQL MERGE INTO upserts, as measured by 29 MERGE-pattern changes and elimination of hash-key null defects, by implementing SCD MERGE logic, deterministic hash-key generation and coalesce-guarded SQL updates.
- Architected the Snowflake consumption layer, as measured by 15 repeatable (R__) Snowflake SQL view migrations across two governed schemas, by designing portfolio, product, share-class and bridge views as a versioned star schema.

- Engineered serverless observability for an Azure Data Factory sync, as measured by 7 pipelines across 4 environments and 50 runs auto-classified, by shipping an Azure Functions, Bicep and PowerShell toolkit with SQL-driven run-history analysis, p95 baselines, severity scoring, deduplicated Jira tickets and Power BI reporting.
- Migrated an analytics platform to vendor API V2, as measured by 12 new V2 tables integrated and the O1–O5 ETL chain validated end to end, by remapping mandate entities, renaming SQL data contracts (tables, columns), hardening configuration and correcting SQL type-coercion defects.
- Modernised the SQL estate behind the platform, as measured by 349 commits touching 1,130 SQL files, 388 stored procedures and 104 staging migrations across 4 release trains, by authoring transaction-safe, idempotent T-SQL migrations (stored procedures, views, functions) automated through YAML pipelines.
- Engineered .NET backend services and RESTful APIs for the reference-data platform, as measured by 1,199 C# files and 24 API controllers across the estate, by building ASP.NET Core services, Entity Framework data-access layers over SQL Server, and publishing product attributes to external consumers through Azure API Management.
- Designed and shipped 17 custom AI agents — ART prompt framework, Copilot Orchestrator, BAU Orchestrator, ADF Sync Agent and self-healing Watchdog, Autonomous Pipeline Monitor (50 runs auto-triaged, MTTD cut to 5 minutes), Database Bottleneck Auditor (SQL-based row/column-level diagnostics isolating 120 mappings and 184 missing registrations), plus Reviewer, Tracker, Teacher, Prompt Crafter and an Agent Builder meta-factory.

Senior Cloud Data Developer / *Janus Henderson Investors*

May 2023 – Sept 2024

- Developed machine-learning institutional opportunity-signal tools analysing \$73B AUM across multi-year horizons, sourcing and shaping features with SQL and PySpark.
- Expanded global investable-universe coverage by 35% using SQL-sourced feature sets and ensemble scoring techniques.
- Built AI agent workflows integrating Snowflake Cortex with Claude and OpenAI via REST APIs and MCP patterns, enabling persona-based GenAI solutions for portfolio management, sales, and institutional reporting over Snowflake SQL semantic layers.

Senior Data Analyst / Data Engineer / *Hubbell Incorporated — Bangalore, India*

Jun 2020 – Aug 2021

- Built Azure Data Factory and Power BI pipelines using T-SQL joins and aggregations to identify unearned cash-discount and deduction opportunities, recovering 15% revenue leakage for the Electronics domain.
- Automated Azure SQL to Power BI data flows using optimized SQL views, accelerating refresh cycles from daily to hourly (80% faster).
- Designed and maintained database objects and stored procedures supporting geospatial and window-function-driven moving-average analytical dashboards, improving decision speed 25%.

Senior Data Analyst / Data Engineer / *Zurich Insurance — Haryana, India*

Dec 2019 – May 2020

- Migrated Oracle SQL (PL/SQL) workloads to T-SQL on SQL Server Management Studio (SSMS)/SSRS hosted on AWS RDS, EC2, S3, and Redshift, enabling 40% faster digital transformation reporting.
- Replaced Autosys batch jobs with Airflow-orchestrated SSIS/Python pipelines running SQL transformations on AWS, cutting overall ETL processing time 35%.
- Configured and managed AWS RDS Oracle instances, EC2 compute, and Redshift SQL warehousing to support enterprise-scale digital transformation and RPA/AI strategy initiatives.

Data Engineer – Credit Risk Modeling / *TransUnion CIBIL Limited — Haryana, India*

Oct 2018 – Nov 2019

- Built Azure Data Factory, Event Hubs, and ADLS Gen2 pipelines for CECL (PD/EAD) survival models, achieving <10% quarterly error rate through rigorous SQL-based data-quality controls, reconciliation queries and automated validation routines.
- Designed segment-level historical loss rate models using reversion-to-mean techniques, translated into production SQL and Python scoring logic; led a 3-analyst team delivering production models 2 weeks ahead of schedule.

Data Analyst / *Citi — Haryana, India*

Feb 2018 – Sept 2018

- Built Decision Tree ML models on SQL-queried transactional credit data to optimize Credit Line Decrease (CLD) policy for one of the largest U.S. credit card issuers, cutting loss rates 20 basis points while improving approval rates from 51% to 90%.
- Developed Python and SAS predictive models forecasting CLD volume, EBIT, and Risk-Adjusted Return (RAR) impacts using SQL-validated transactional datasets, enabling multi-million dollar policy optimization decisions.

Data Engineer / Fraud Analyst / *American Express — Haryana, India*

Jan 2017 – Jan 2018

- Built Spark, Spark SQL, Kafka, and Airflow real-time pipelines processing high-volume transactional data for fraud detection, reducing data latency 40% and enabling near real-time AI scoring.
- Built KModes clustering and stochastic gradient descent (SGD) models in Python and SAS on SQL-extracted feature sets for fraud pattern detection, improving detection accuracy 18% through iterative feature engineering.
- Designed SQL-based data-quality and cleansing rules (pattern-matching queries) identifying disposable email patterns, blocking 25% of synthetic account applications before downstream fraud review.

- Won the American Express Data Science Championship for cardmember characteristics and risky transaction detection pattern handbooks.

Data Engineer / Associate Data Scientist / *Cognizant — Bangalore, India*

Jun 2016 – Jan 2017

- Built AWS EC2, S3, and Step Functions ETL pipelines orchestrating SQL-based transformations and automating 85% of data workflows, reducing manual processing effort and improving reliability of nightly batch jobs.
- Deployed 5 machine learning models (Logistic Regression, Random Forest, SVM, ANN, Decision Trees) in R and Python on SQL-extracted, feature-engineered datasets, achieving 92% prediction accuracy versus a 78% baseline.
- Built data pipelines leveraging AWS VPC, ELB, Auto Scaling Groups, Route 53, IAM, CloudTrail, CloudWatch, CloudFormation, CloudFront, SNS, and RDS SQL databases.
- Created an IBM Watson NLP chatbot using the Conversation API to optimize IT ticket allocation, reducing resolution time 30%.
- Worked with Snowflake cloud data warehouse and AWS S3 to integrate multiple source systems, using Snowflake SQL (VARIANT, FLATTEN) for nested JSON ingestion into downstream analytics tables.

AI AGENT PORTFOLIO

- **ART prompt framework** — 132-line multi-agent standard across 2 repos; eliminates ungrounded AI output.
- **Copilot Orchestrator** — swarm orchestrator automating ADO/ADF triage; one entry point for cross-system incidents.
- **BAU Orchestrator** — BAU triage, evidence capture and RCA (SQL evidence queries); underpinned 9 documented problem themes.
- **ADF Sync Agent & Watchdog** — triages run histories and database error logs via T-SQL diagnostics, self-heals an hourly pipeline where a missed run loses that hour of data.
- **Autonomous Pipeline Monitor** — 94 files, 7 pipelines, 4 environments; 50 runs auto-triaged, detection cut to 5 minutes.
- **Database Bottleneck Auditor** — SQL row/column-level diagnostics; isolated 120 HFD mappings and 184 missing registrations.
- **Autonomous Software Engineer** — requirements to ASP.NET and SQL changes (stored procedures, views, migrations) with impact traced before code ships.
- **Tracker / Reviewer / Expert Loop Engineer** — correlates failures to precedent, QA-gates code and SQL scripts, and runs recursive fix-test-refix cycles before release.
- **Context Ingestion / File Reader / Token Optimizer** — builds workspace models from ADO/git history, extracts context from PDFs, logs and SQL files, and compresses payloads before model calls.
- **Teacher / Prompt Crafter / Agent Builder** — converts incidents into Confluence post-mortems, standardises agent prompts, and scaffolds new specialist agents in minutes.
- **API Migration Validator** — validates vendor API migrations across 4 environments with SQL reconciliation checks; proved a 12-table V2 cutover.

EDUCATION

- Doctor of Business Administration | Belhaven University | 2025 – Present
- Master of Science, Management Sciences and Quantitative Data Analytics | Mount Vernon Nazarene University | 2024 – 2025
- Master of Science, Applied Computer Science | Northwest Missouri State University | 2021 – 2022
- Master of Technology (M.Tech) & Bachelor of Technology (B.Tech), Mechanical Engineering (Intelligent Manufacturing) | Indian Institute of Technology (IIT) Madras | 2011 – 2016

CERTIFICATIONS & AWARDS

- Google AI Professional Certificate | Google
- Academy Accreditation – Generative AI Fundamentals | Databricks
- AWS Cloud Practitioner Essentials | Amazon Web Services (AWS)
- Big Data Foundations – Level 1 | IBM
- Certified Data Scientist – INSOFE (2016)
- American Express Data Science Championship Winner (2017)
- Janus Henderson Quarterly Town Hall Shout Out Award